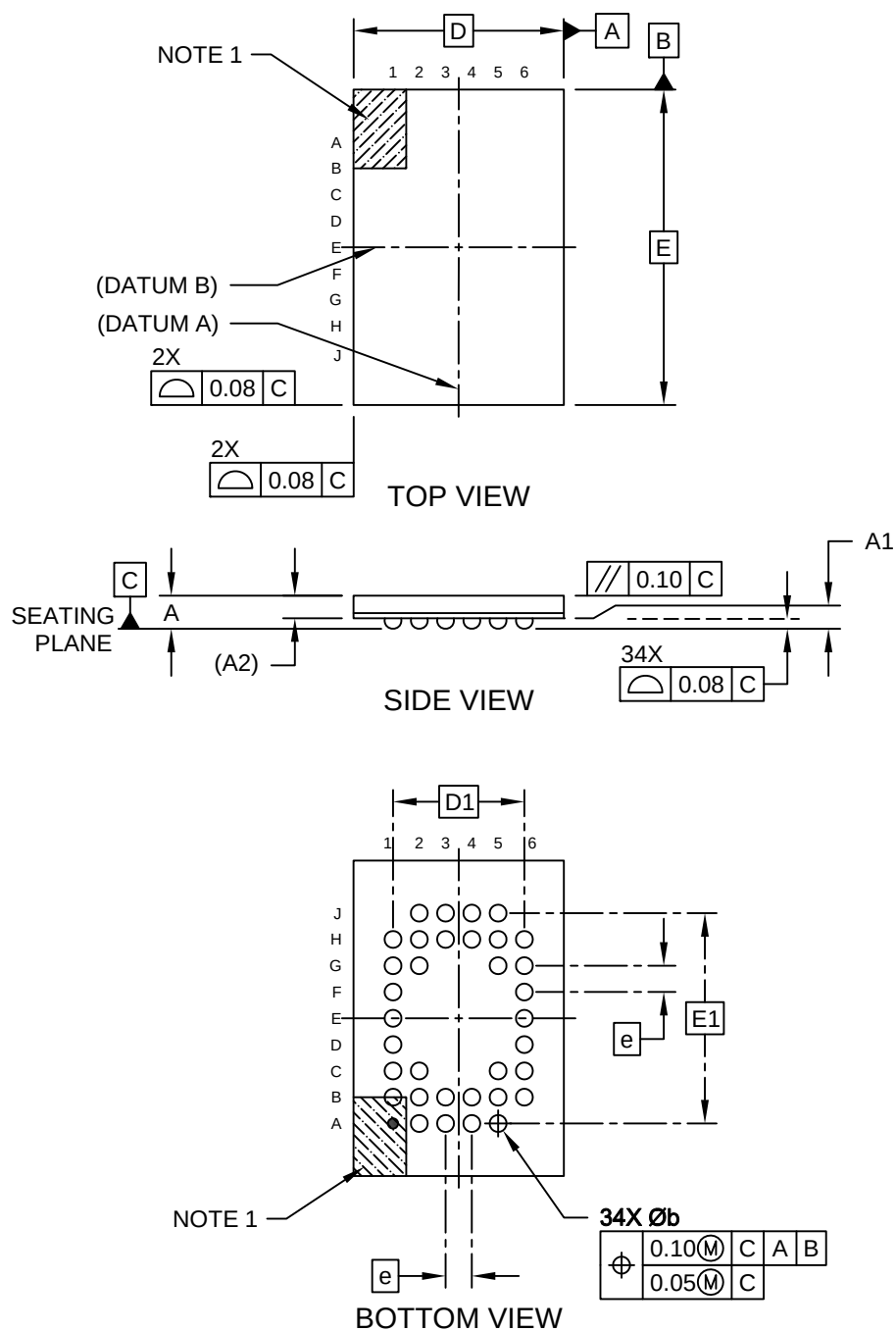


## Packaging Diagrams and Parameters

### 34-Ball Very, Very Thin Fine Pitch Ball Grid Array (CP) - 4x6 mm Body [WFBGA] Supertex Legacy Packages MAME and MAMF

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

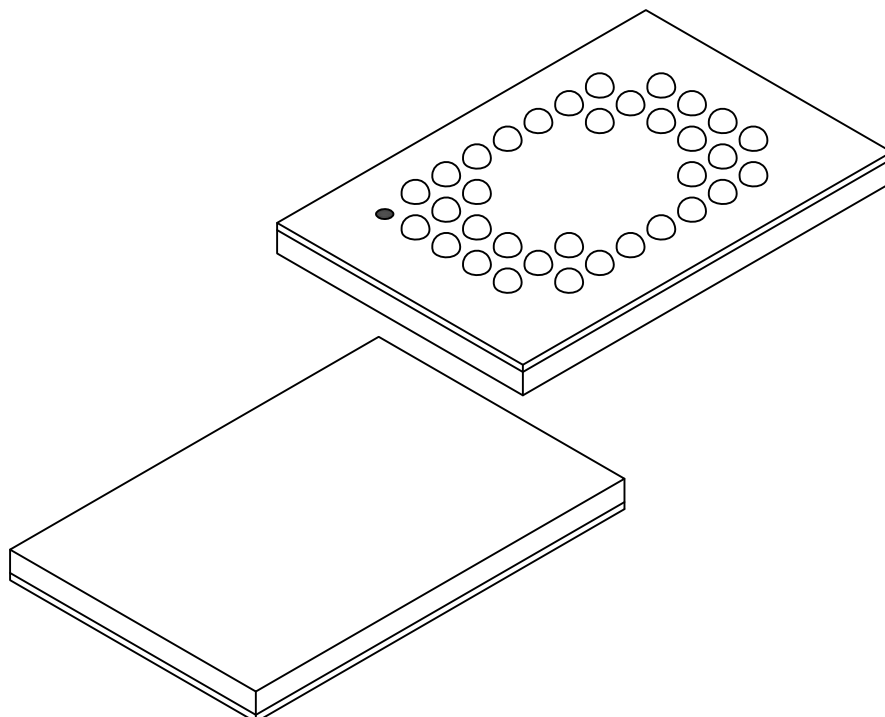


Microchip Technology Drawing C04-164-CP Rev A Sheet 1 of 2

## Packaging Diagrams and Parameters

### 34-Ball Very, Very Thin Fine Pitch Ball Grid Array (CP) - 4x6 mm Body [WFBGA] Supertex Legacy Packages MAME and MAMF

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



|                          |    | Units | MILLIMETERS |          |      |
|--------------------------|----|-------|-------------|----------|------|
| Dimension Limits         |    |       | MIN         | NOM      | MAX  |
| Number of Terminals      | N  |       | 34          |          |      |
| Pitch                    | e  |       | 0.50 BSC    |          |      |
| Overall Height           | A  |       | 0.53        | 0.63     | 0.73 |
| Standoff                 | A1 |       | 0.14        | 0.20     | 0.26 |
| Package Thickness        | A2 |       | 0.43 REF    |          |      |
| Overall Length           | D  |       | 4.00 BSC    |          |      |
| Overall Terminal Spacing | D1 |       |             | 2.50 BSC |      |
| Overall Width            | E  |       | 6.00 BSC    |          |      |
| Exposed Pad Width        | E2 |       |             | 4.00 BSC |      |
| Terminal Width           | b  |       | 0.27        | 0.32     | 0.37 |

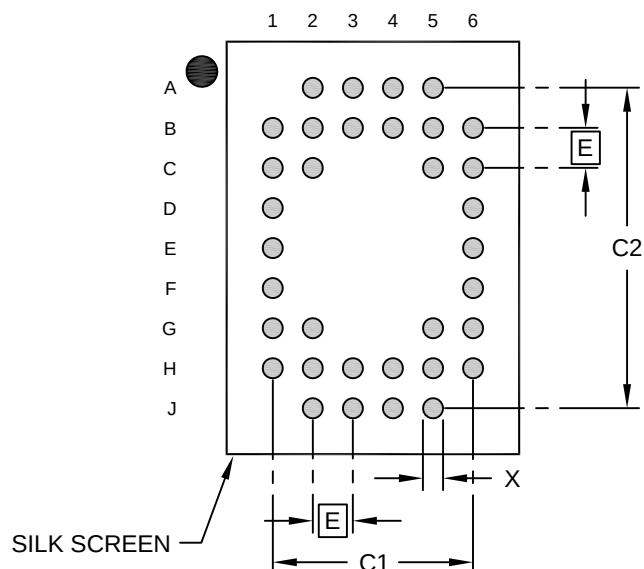
**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, for information purposes only.

## Land Pattern

### 34-Ball Very, Very Thin Fine Pitch Ball Grid Array (CP) - 4x6 mm Body [WFBGA] Supertex Legacy Packages MAME and MAMF

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



### RECOMMENDED LAND PATTERN

| Units                   |    | MILLIMETERS |      |      |
|-------------------------|----|-------------|------|------|
| Dimension Limits        |    | MIN         | NOM  | MAX  |
| Contact Pitch           | E  | 0.50 BSC    |      |      |
| Contact Pad Spacing     | C1 |             | 2.50 |      |
| Contact Pad Spacing     | C2 |             | 4.00 |      |
| Contact Pad Width (X34) | X  |             |      | 0.25 |

#### Notes:

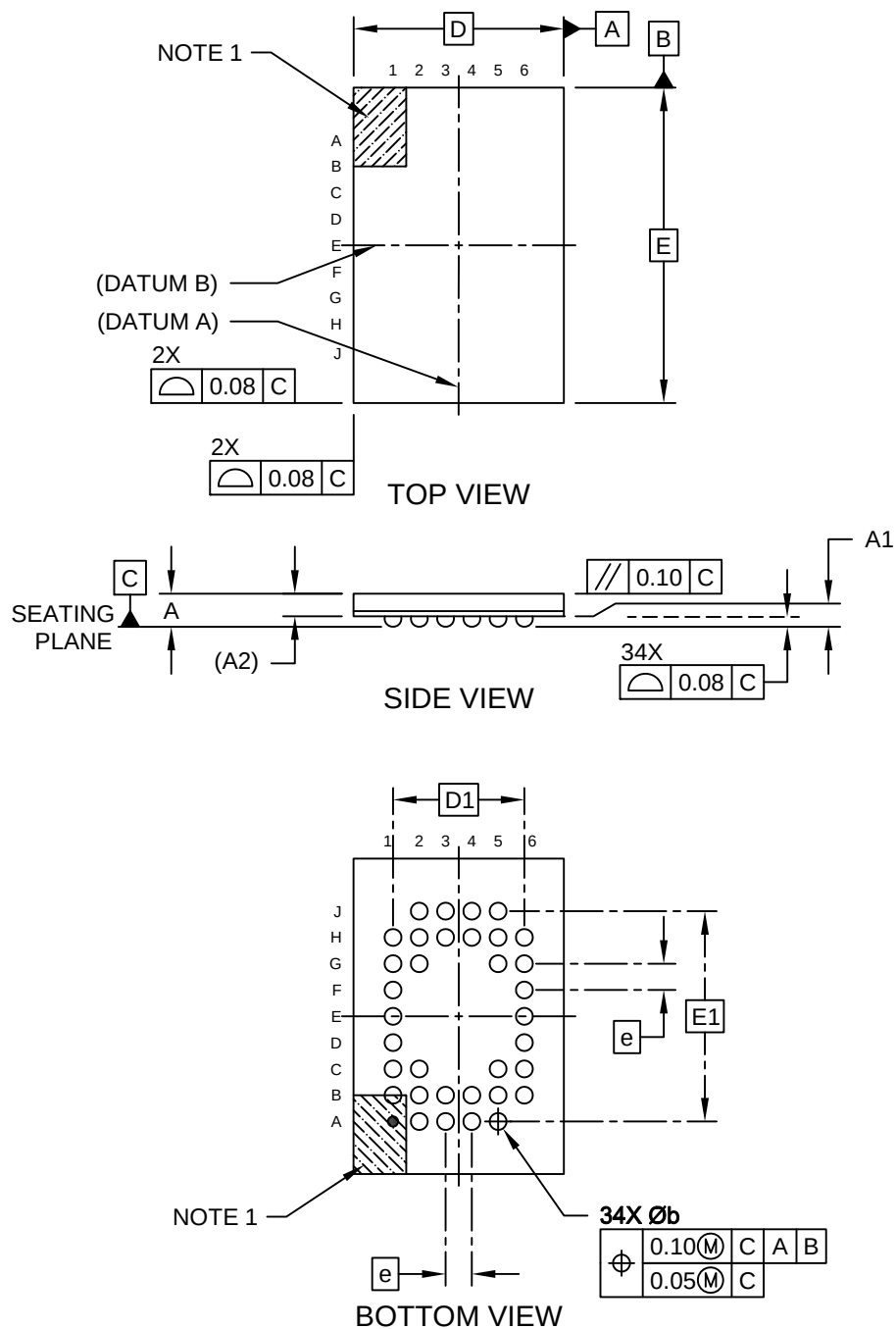
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2164-CP Rev A

## Packaging Diagrams and Parameters

### 34-Ball Very, Very Thin Fine Pitch Ball Grid Array (CU) - 4x6 mm Body [WFBGA] Supertex Legacy Packages MME and MMF

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

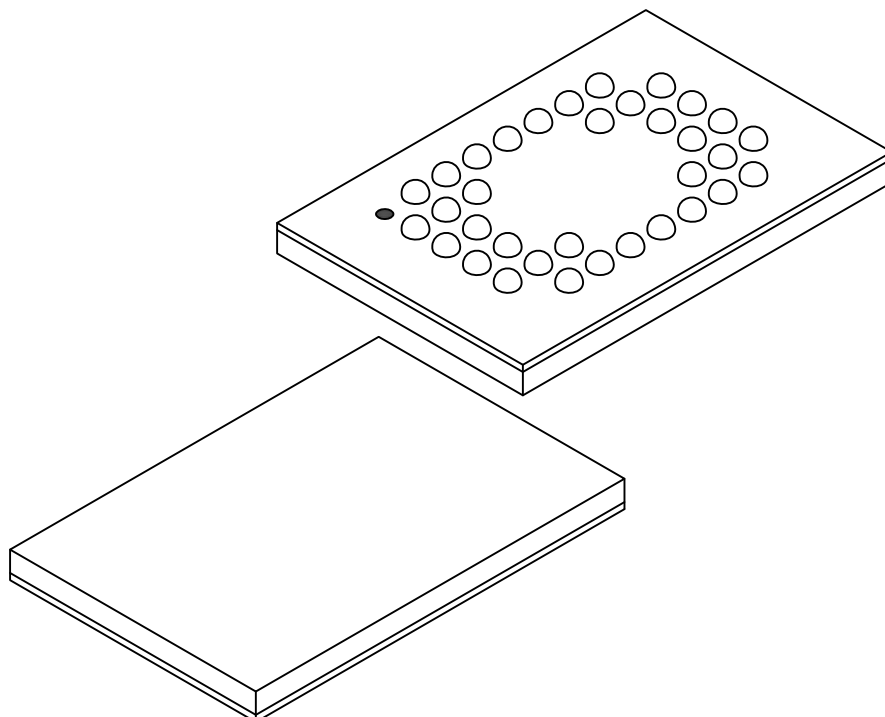


Microchip Technology Drawing C04-164-CU Rev A Sheet 1 of 2

## Packaging Diagrams and Parameters

### 34-Ball Very, Very Thin Fine Pitch Ball Grid Array (CU) - 4x6 mm Body [WFBGA] Supertex Legacy Packages MME and MMF

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



|                          |    | Units | MILLIMETERS |          |      |
|--------------------------|----|-------|-------------|----------|------|
| Dimension Limits         |    |       | MIN         | NOM      | MAX  |
| Number of Terminals      | N  |       | 34          |          |      |
| Pitch                    | e  |       | 0.50 BSC    |          |      |
| Overall Height           | A  |       | 0.53        | 0.63     | 0.73 |
| Standoff                 | A1 |       | 0.14        | 0.20     | 0.26 |
| Package Thickness        | A2 |       | 0.43 REF    |          |      |
| Overall Length           | D  |       | 4.00 BSC    |          |      |
| Overall Terminal Spacing | D1 |       |             | 2.50 BSC |      |
| Overall Width            | E  |       | 6.00 BSC    |          |      |
| Exposed Pad Width        | E2 |       |             | 4.00 BSC |      |
| Terminal Width           | b  |       | 0.27        | 0.32     | 0.37 |

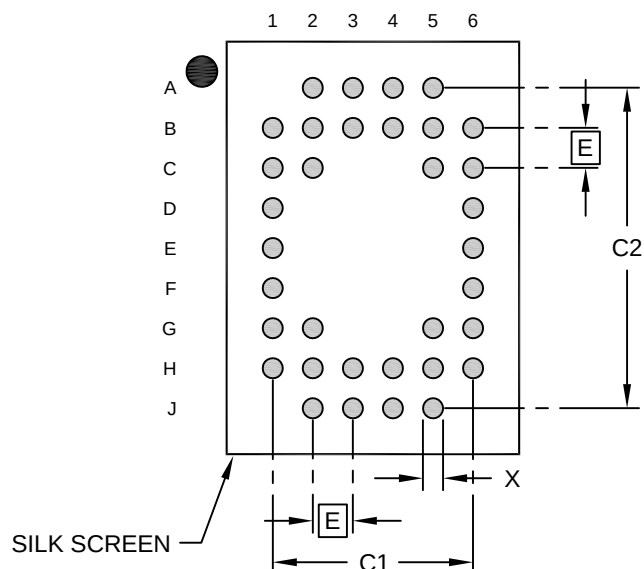
**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, for information purposes only.

## Land Pattern

### 34-Ball Very, Very Thin Fine Pitch Ball Grid Array (CU) - 4x6 mm Body [WFBGA] Supertex Legacy Packages MME and MMF

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



### RECOMMENDED LAND PATTERN

| Units                   |    | MILLIMETERS |      |      |
|-------------------------|----|-------------|------|------|
| Dimension Limits        |    | MIN         | NOM  | MAX  |
| Contact Pitch           | E  | 0.50 BSC    |      |      |
| Contact Pad Spacing     | C1 |             | 2.50 |      |
| Contact Pad Spacing     | C2 |             | 4.00 |      |
| Contact Pad Width (X34) | X  |             |      | 0.25 |

#### Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2164-CU Rev A